

Ilian Khankhalaev

Vancouver, BC, Canada

☎ 672-272-1832 ✉ ika34@sfu.ca [in LinkedIn](#) [@ GitHub](#) [🌐 Portfolio](#)

Education

Simon Fraser University

Bachelor of Applied Science in Computer Science & Minor in Mathematics | GPA: 3.77

Jan. 2024 — Dec. 2027 (Expected)

Burnaby, BC

Experience

Research Software Contributor

Jul. 2026 — Present

Simon Fraser University | Computational Geometry & Biomechanical Simulation Research

Burnaby, BC

- Developing a Python-based **skin-to-muscle registration pipeline** for 16k+ vertex anatomical meshes, leveraging Signed Distance Functions (SDFs), computational geometry, and Autodesk Maya to automate biomechanical modeling workflows.
- Implementing and evaluating mesh registration, deformation, and **collision-handling algorithms** across 43 anatomical structures, utilizing K-nearest surface blending (K=5), Laplacian smoothing, and optimization techniques for anatomically accurate skin attachment.

Software Engineering Intern

Mar. 2025 — Dec. 2025

SYNKRON | Data Infrastructure & Platform Team

Miami, FL (Remote)

- Developed **Python backend automation** tools for ingestion, transformation, and validation of structured datasets, improving processing efficiency across internal workflows.
- Implemented **SQL-based validation** scripts and integrity checks to identify schema anomalies and ensure consistency across relational databases.
- Integrated **Azure Storage** with cloud processing pipelines to support reliable data staging, downstream analytics, and automated processing workflows.

Operations Assistant

May 2021 — Aug. 2023

Khankhalaev Gallery | Operations & Logistics Management

Moscow, Russia

- Improved inventory data reliability across 400+ art assets by implementing spreadsheet **validation checks** and maintaining pricing, logistics, and inventory records in Excel and Google Sheets.
- Enhanced operational visibility for exhibitions and sales activities by producing tracking reports and consolidating data from multiple business functions into centralized summaries.

Technical Projects

AegisOps [🔗](#) | C#, ASP.NET Core, React, TypeScript, Python, Azure OpenAI, Azure, Docker

Jul. 2026

Microsoft AI Hackathon Grand Champion (Overall Best Project @ Microsoft HQ)

- Engineered an **AI-driven cloud operations platform** for 49 Azure regions, enabling proactive workload redistribution through real-time telemetry.
- Built a **geospatial infrastructure model** by mapping 34,000+ power plants to nearby **Azure regions**, enabling sustainability-aware workload planning.
- Implemented GPT-4o-powered decision-making over live telemetry to recommend **cross-region workload redistribution** based on latency, cost, carbon intensity, and service availability.

DataLy [🔗](#) | AWS, Java, FastAPI, Next.js, PySpark, PostgreSQL, Kafka, Airflow, Python, Gemini API, Docker

May 2026 — Present

- Deployed a cloud-native analytics platform on **AWS** using 7+ containerized services, provisioning scalable FastAPI APIs for batch analytics.
- Engineered an event-driven **data ingestion pipeline** using Kafka to coordinate asynchronous PySpark workers, processing 1,200+ row datasets across 13-column schemas for automated profiling datasets into 6+ statistical summaries and visualizations.
- Developed a context-aware analytics chatbot integrating the **Gemini API** with a 12-message sliding memory window, enabling conversational querying and automated insights over dataset profiles.

ArbScanner [🔗](#) | FastAPI, Vite, React, MySQL, Python, Gemini API, Vercel, Render, Railway

Mar. 2026 — Apr. 2026

- Engineered a deterministic **arbitrage engine** to identify semantically equivalent prediction markets across 3 exchanges (Polymarket, Manifold, Kalshi), leveraging Gemini-assisted title matching.
- Built a local-first operator workstation with a **FastAPI backend**, React frontend, and managed MySQL database to stage real-time event snapshots.

Leadership & Mentorship

Calculus Mentor

Jan. 2026 — Apr. 2026

Simon Fraser University | Calculus & Foundational Mathematics

Burnaby, BC

- Mentored breakout groups of **8–10** calculus students, guiding problem-solving sessions in coordination with course Teaching Assistants.
- Led mathematical discussions across **20+ seminar** sessions to improve student analytical reasoning and clarify complex concepts.

Certifications & Awards

Awards: Microsoft AI Hackathon Grand Champion (Overall Best Project @ Microsoft HQ), Dean's Honour Roll (Fall 2024, Spring 2026)

Certifications: AWS Certified Cloud Practitioner (CLF-C02), Supervised Machine Learning (Stanford University), Claude Code Certified

Technical Skills

Programming Languages: Python, Java, C#, SQL, JavaScript, C/C++, R, HTML, CSS

Cloud & Data Engineering: AWS (EC2, S3), Azure (Storage, OpenAI), Apache Spark, Apache Kafka, Docker, ETL, Data Pipelines, Apache Airflow

Frameworks & Technologies: ASP.NET Core, FastAPI, Flask, React, Next.js, Vite, React Router, Leaflet, REST APIs

Databases: PostgreSQL, MySQL, SQLite

Developer Tools: Git/GitHub, GitHub Actions, Render, Railway, Vercel, VS Code, Jupyter Notebook, Tableau

Libraries: Pandas, NumPy, scikit-learn, CatBoost, Matplotlib, Seaborn